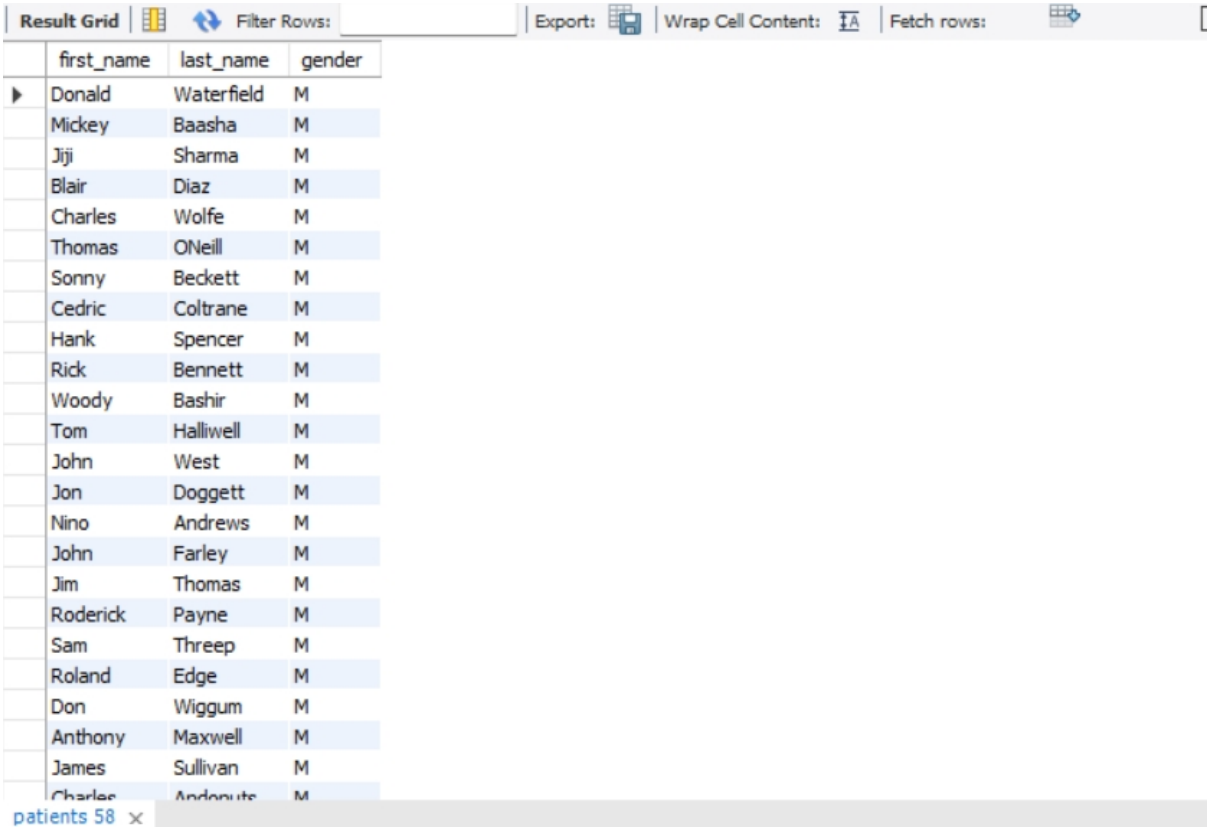
PRSQ-01-Medical Data History

Perform the Problem Queries:

1. Show first name, last name, and gender of patients who's gender is 'M'

Q-select first_name,last_name,gender from patients where gender='M';



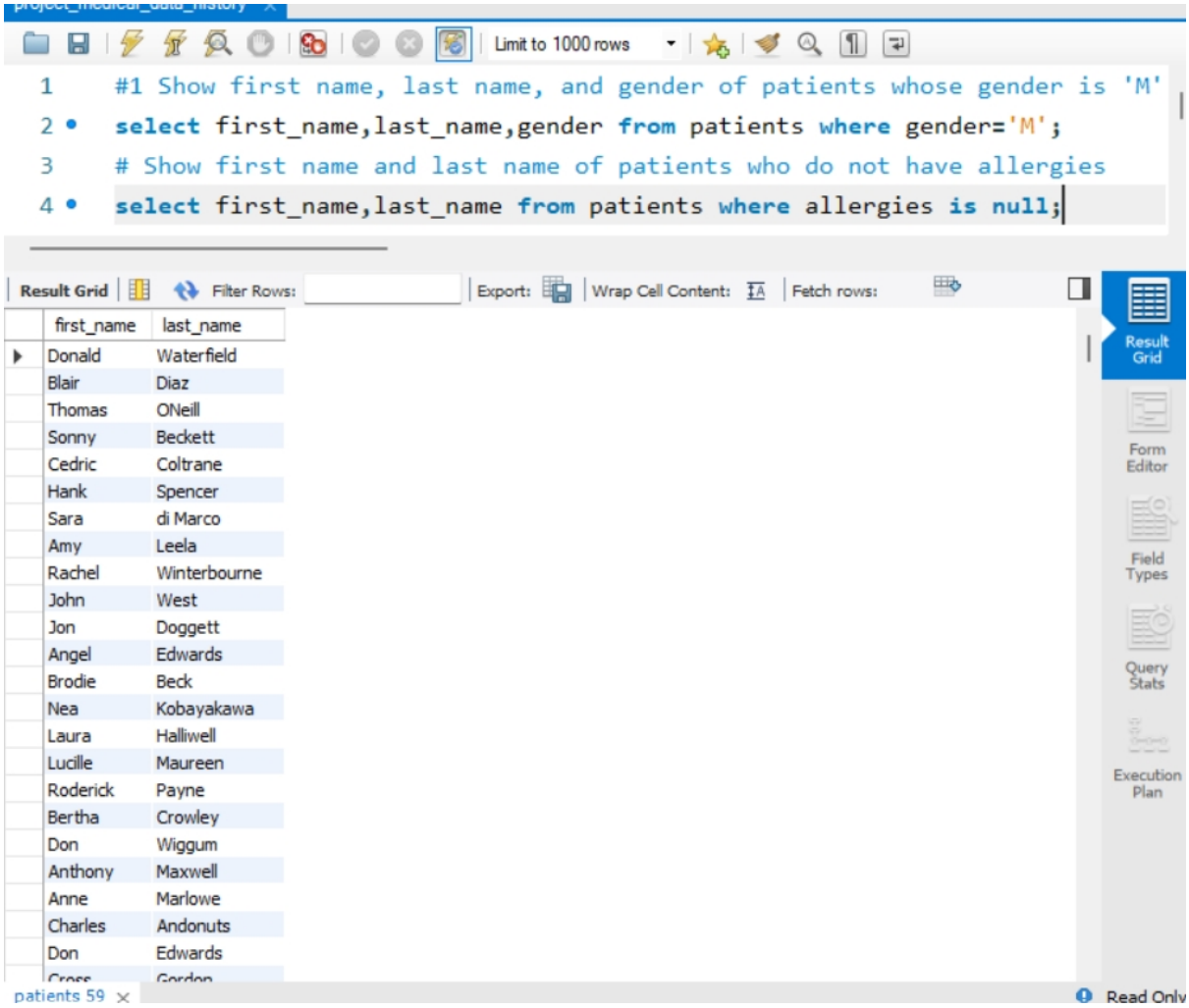
Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

first_name	last_name	gender
Donald	Waterfield	M
Mickey	Baasha	M
Jiji	Sharma	M
Blair	Diaz	M
Charles	Wolfe	M
Thomas	ONeill	M
Sonny	Beckett	M
Cedric	Coltrane	M
Hank	Spencer	M
Rick	Bennett	M
Woody	Bashir	M
Tom	Halliwell	M
John	West	M
Jon	Doggett	M
Nino	Andrews	M
John	Farley	M
Jim	Thomas	M
Roderick	Payne	M
Sam	Threep	M
Roland	Edge	M
Don	Wiggum	M
Anthony	Maxwell	M
James	Sullivan	M
Charles	Andonuts	M

patients 58 ×

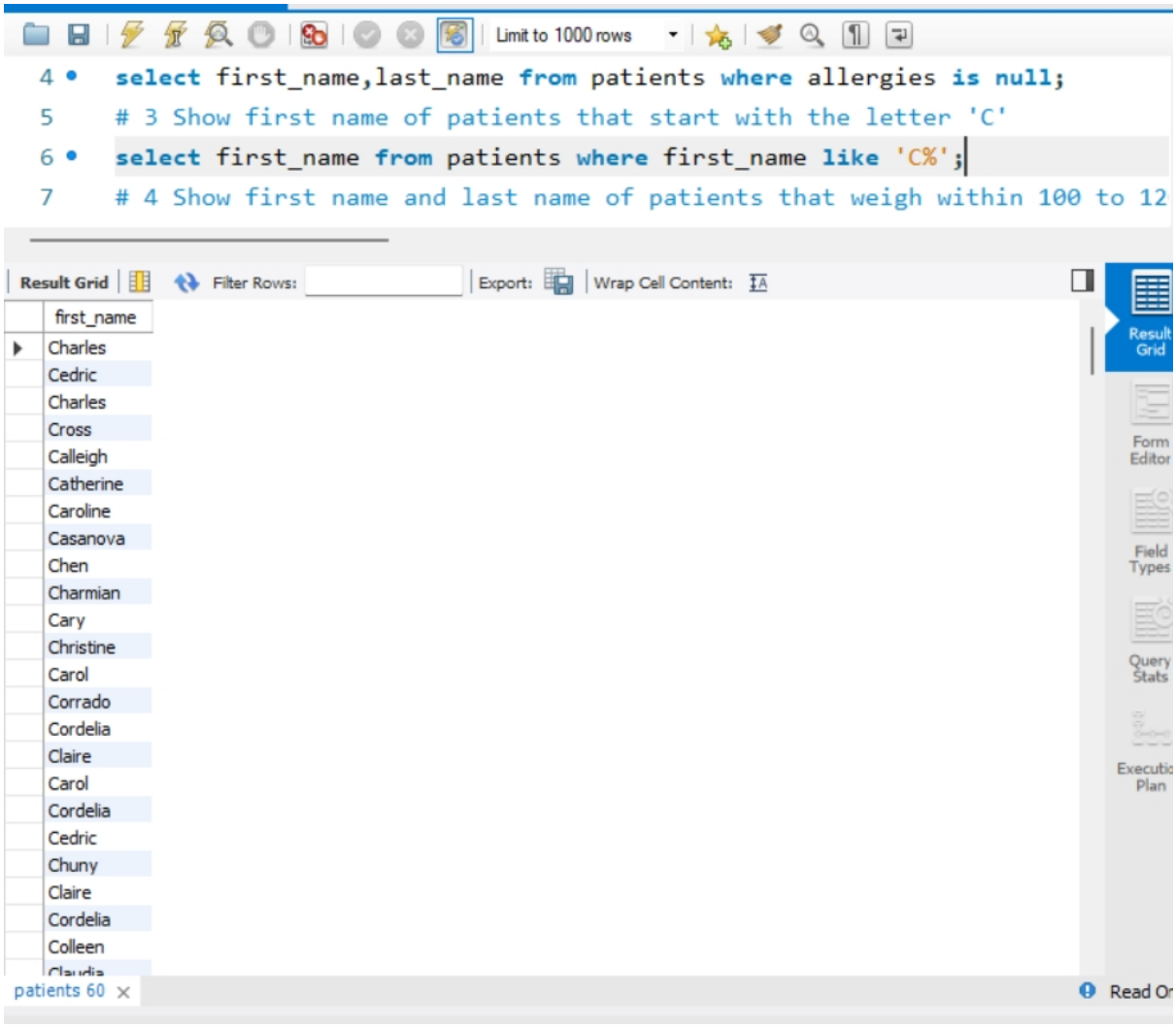
2. Show first name and last name of patients who does not have allergies.

Q--select first_name,last_name from patients where allergies is null;



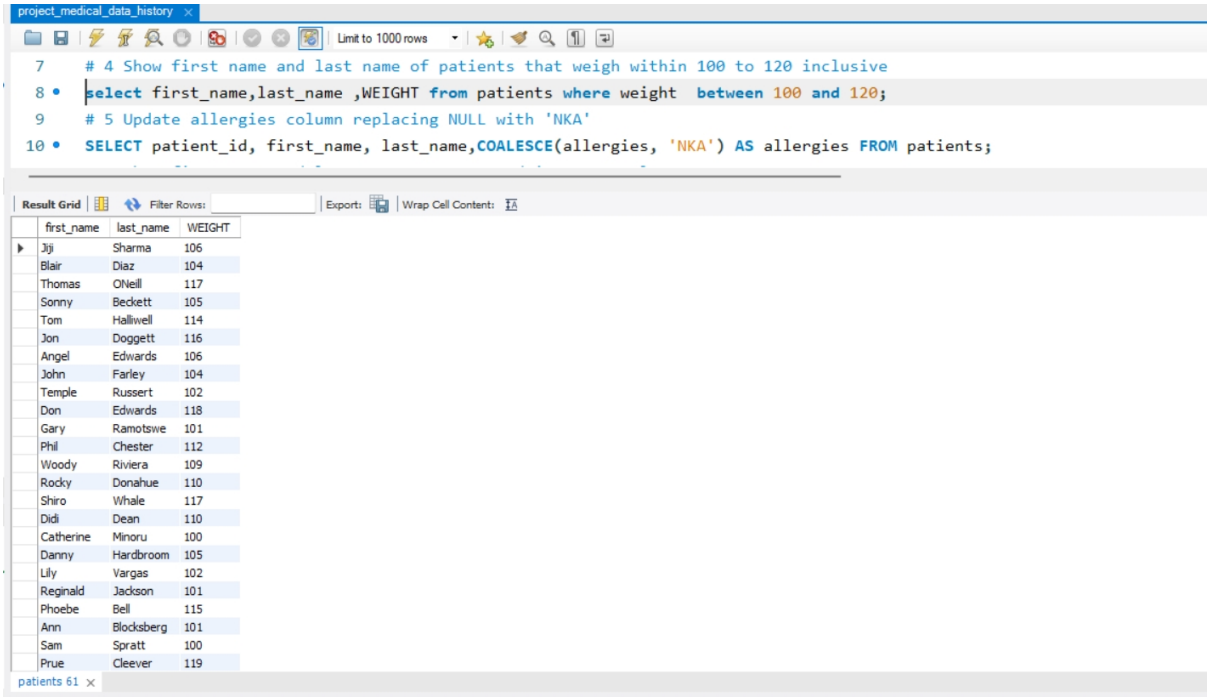
3. Show first name of patients that start with the letter 'C'

Q--select first_name from patients where first_name like 'C%';



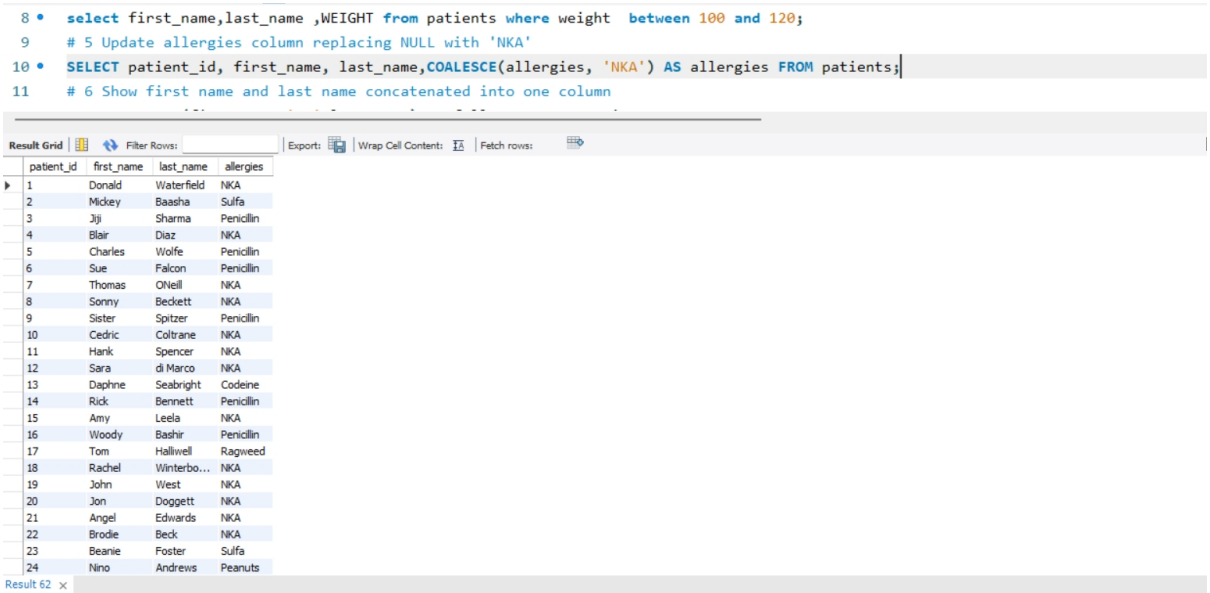
4. Show first name and last name of patients that weight within the range of 100 to 120 (inclusive)

Q--select first_name,last_name ,WEIGHT from patients where weight between 100 and 120;



5. Update the patients table for the allergies column. If the patient's allergies is null then replace it with 'NKA'

Q--SELECT patient_id, first_name, last_name,COALESCE(allergies, 'NKA') AS allergies FROM patients;



6. Show first name and last name concatenated into one column to show their full name.

Q-- select concat(first_name,' ',last_name) as full_name from patients;

```
11    # 6 Show first name and last name concatenated into one column
12 •  select concat(first_name,'  ',last_name) as full_name from patients;
13    -- 7 Show first name, last name, and full province name of each patient
14 •  select p.first_name,p.last_name,pr.province_name from patients p join province_names pr on p.province_id=pr.province_id;
```

full_name
Donald Waterfield
Mickey Baasha
Jiji Sharma
Blair Diaz
Charles Wolfe
Sue Falcon
Thomas ONeill
Sonny Beckett
Sister Spitzer
Cedric Coltrane
Hank Spencer
Sara di Marco
Daphne Seabright
Rick Bennett
Amy Leela
Woody Bashir
Tom Halliwell
Rachel Winterbo...
John West
Jon Doggett
Angel Edwards
Brodie Beck
Beanie Foster
Nino Andrews

7. Show first name, last name, and the full province name of each patient.

Q--select p.first_name,p.last_name,pr.province_name from patients p join province_names pr on p.province_id=pr.province_id;

```
13    -- 7 Show first name, last name, and full province name of each patient
14 •  select p.first_name,p.last_name,pr.province_name from patients p join province_names pr on p.province_id=pr.province_id;
15    -- 8 Show count of patients born in 2010
16 •  select count(*) from  patients where year( birth_date)=2010;
```

first_name	last_name	province_name
Donald	Waterfield	Ontario
Mickey	Baasha	Ontario
Jiji	Sharma	Ontario
Blair	Diaz	Ontario
Charles	Wolfe	Ontario
Sue	Falcon	Ontario
Thomas	ONeill	Ontario
Sonny	Beckett	Nova Scotia
Sister	Spitzer	Ontario
Cedric	Coltrane	Ontario
Hank	Spencer	Ontario
Sara	di Marco	Ontario
Daphne	Seabright	Ontario
Rick	Bennett	Ontario
Amy	Leela	Ontario
Woody	Bashir	Ontario
Tom	Halliwell	Ontario
Rachel	Winterbo...	Ontario
John	West	Ontario
Jon	Doggett	Ontario
Angel	Edwards	Ontario
Brodie	Beck	Ontario
Beanie	Foster	Ontario
Nino	Andrews	Ontario

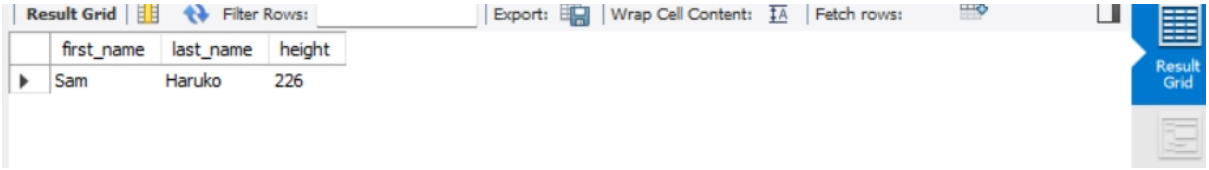
8. Show how many patients have a birth_date with 2010 as the birth year.

Q--select count(*) from patients where year(birth_date)=2010;

count(*)
55

9. Show the first_name, last_name, and height of the patient with the greatest height.

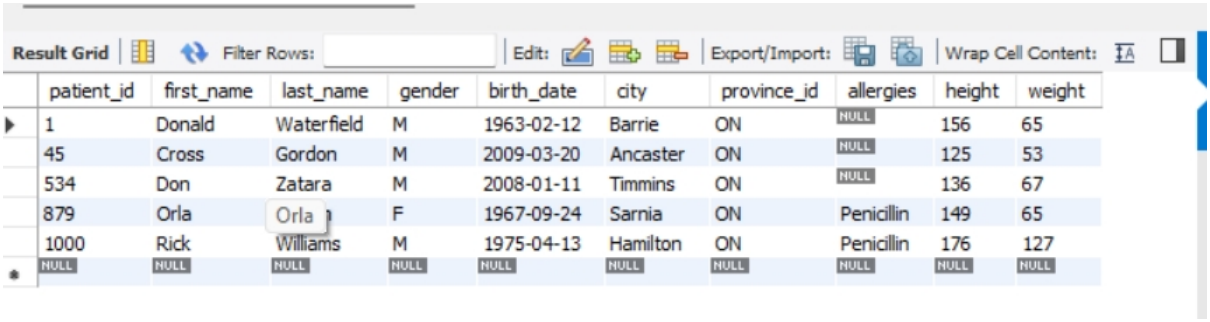
Q--select first_name,last_name,height from patients order by height desc limit 1;



first_name	last_name	height
Sam	Haruko	226

10. Show all columns for patients who have one of the following patient_ids: 1,45,534,879,1000

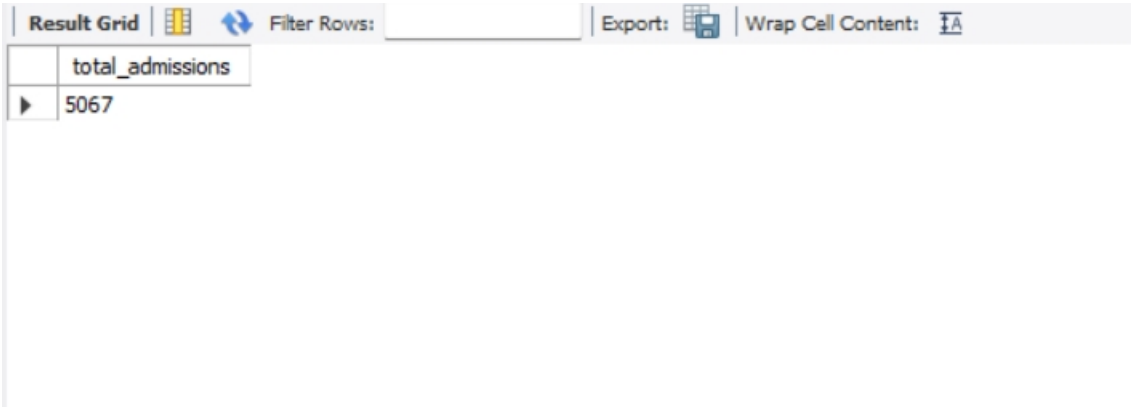
Q--select * from patients where patient_id in(1,45,534,879,1000);



patient_id	first_name	last_name	gender	birth_date	city	province_id	allergies	height	weight
1	Donald	Waterfield	M	1963-02-12	Barrie	ON	NULL	156	65
45	Cross	Gordon	M	2009-03-20	Ancaster	ON	NULL	125	53
534	Don	Zatara	M	2008-01-11	Timmins	ON	NULL	136	67
879	Orla	Orla	F	1967-09-24	Sarnia	ON	Penicillin	149	65
1000	Rick	Williams	M	1975-04-13	Hamilton	ON	Penicillin	176	127
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

11. Show the total number of admissions

Q--select count(*) as total_admissions from admissions;



total_admissions
5067

12. Show all the columns from admissions where the patient was admitted and discharged on the same day.

Q--select * from admissions where admission_date=discharge_date;

```
22 •  select count(*) as total_admissions from admissions;
23    -- 12 Show admissions where admitted and discharged same day
24 •  select * from admissions where admission_date=discharge_date;
25    -- 13 Show total admissions for patient_id 579
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

patient_id	admission_date	discharge_date	diagnosis	attending_doctor_id
1	2018-09-20	2018-09-20	Ineffective Breathin Pattern R/T Fluid Accumulatio	24
9	2018-12-31	2018-12-31	Ruptured Appendicitis	19
10	2019-02-27	2019-02-27	Lower Quadrant Pain	27
17	2019-03-04	2019-03-04	Diabetes Mellitus	9
28	2019-03-30	2019-03-30	Cancer Of The Stomach	26
31	2018-09-26	2018-09-26	Cardiovascular Disease	19
53	2018-10-24	2018-10-24	Urinary Tract Infection	8
54	2019-04-07	2019-04-07	Hypertension	21
70	2018-07-17	2018-07-17	Migraine	20
78	2018-06-17	2018-06-17	Hypertension	17
91	2018-08-30	2018-08-30	Congestive Heart Failure	3
92	2019-01-03	2019-01-03	Osteo Arthitis Knee	5
93	2019-05-20	2019-05-20	Left Creebral Vascular Accident	12
110	2019-02-18	2019-02-18	Biliary Colic	14
111	2018-08-25	2018-08-25	Subdural Hematoma	27
113	2018-10-21	2018-10-21	2nd & 3rd Degree Burns To Right Arm / Both Legs	13
114	2019-02-11	2019-02-11	MVA HI	26
120	2018-10-04	2018-10-04	Uterine Fibroid	26
123	2019-03-30	2019-03-30	Pregnancy	6
131	2018-06-07	2018-06-07	Amyotopic Lateral Sdetosis	20
132	2018-07-03	2018-07-03	Appendicitis	2
145	2018-10-30	2018-10-30	Rll Pneumonia	20
151	2019-01-23	2019-01-23	Broken Hip	27
156	2018-09-11	2018-09-11	Scatica	11

admissions 66 ×

13. Show the total number of admissions for patient_id 579.

Q--select count(*)total_admissions from admissions where patient_id=579;

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

total_admissions
2

14. Based on the cities that our patients live in, show unique cities that are in province_id 'NS'?

Q--select distinct city from patients where province_id='NS';

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

city
Port Hawkesbury
Halifax
Inverness

15. Write a query to find the first_name, last name and birth date of patients who have height more than 160 and weight more than 70

Q--select first_name,last_name,birth_date from patients where height>160 and weight>70;

```
28 • select distinct city from patients where province_id='NS';
29   -- 15 Show first name, last name, birth date for patients with height>160 and weight>70
30 • select first_name,last_name,birth_date  from patients where height>160 and weight>70;
31   -- 16 Show unique birth years ordered ascending
```

first_name	last_name	birth_date
Mickey	Baasha	1981-05-28
Jiji	Sharma	1957-09-05
Blair	Diaz	1967-01-07
Thomas	ONeill	1993-01-31
Sonny	Beckett	1952-12-11
Sister	Spitzer	1966-10-15
Rick	Bennett	1977-01-27
Amy	Leela	1977-06-25
Tom	Halliwell	1987-08-01
Rachel	Winterbourne	1966-04-26
Jon	Doggett	1951-12-25
Angel	Edwards	1975-08-22
Nino	Andrews	2001-04-21
John	Farley	1956-09-12
Lucille	Maureen	1953-08-22
Sam	Threep	1982-06-16
Roland	Edge	1967-09-25
Don	Wiggum	2002-11-15
Anne	Marlowe	1996-09-17
Temple	Russert	1953-05-11
Don	Edwards	1975-07-02
Neil	Allan	1975-02-02
Harry	Kuramitsu	1979-12-03
Gary	Ramotswe	1977-07-31

16. Show unique birth years from patients and order them by ascending.

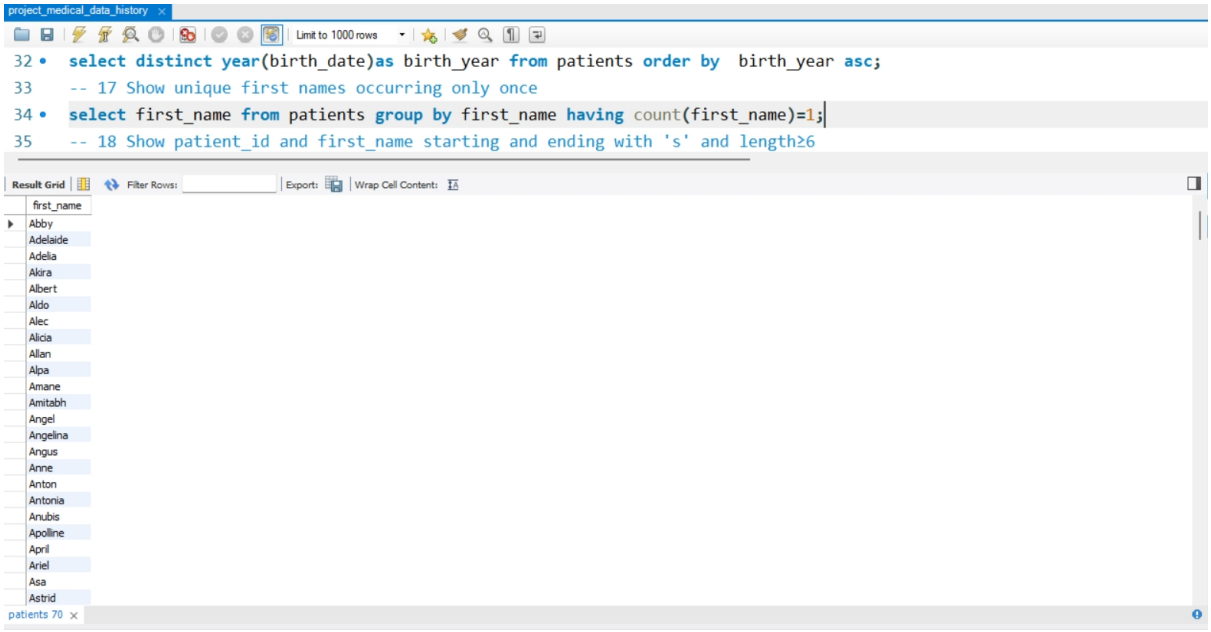
Q--select distinct year(birth_date)as birth_year from patients order by birth_year asc;

```
31   -- 16 Show unique birth years ordered ascending
32 • select distinct year(birth_date)as birth_year from patients order by  birth_year asc;
33   -- 17 Show unique first names occurring only once
34 • select first_name from patients group by first_name having count(first_name)=1;
```

birth_year
1918
1923
1925
1926
1927
1928
1929
1931
1933
1934
1936
1937
1938
1939
1940
1941
1942
1943
1944
1945
1946
1947
1948
1949

17. Show unique first names from the patients table which only occurs once in the list.

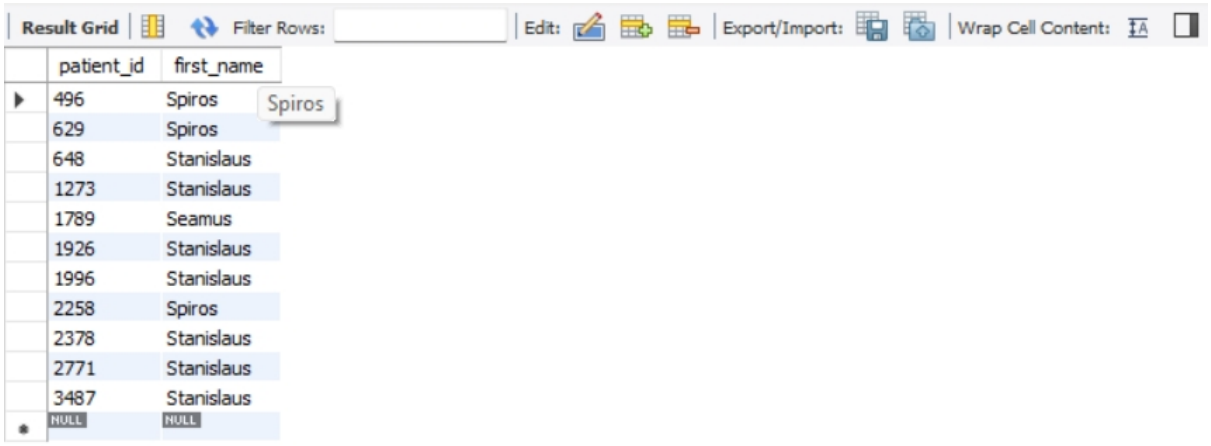
For example, if two or more people are named 'John' in the first_name column then don't include their name in the output list. If only 1 person is named 'Leo' then include them in the output. Tip: HAVING clause was added to SQL because the WHERE keyword cannot be used with aggregate functions.



Q--select first_name from patients group by first_name having count(first_name)=1;

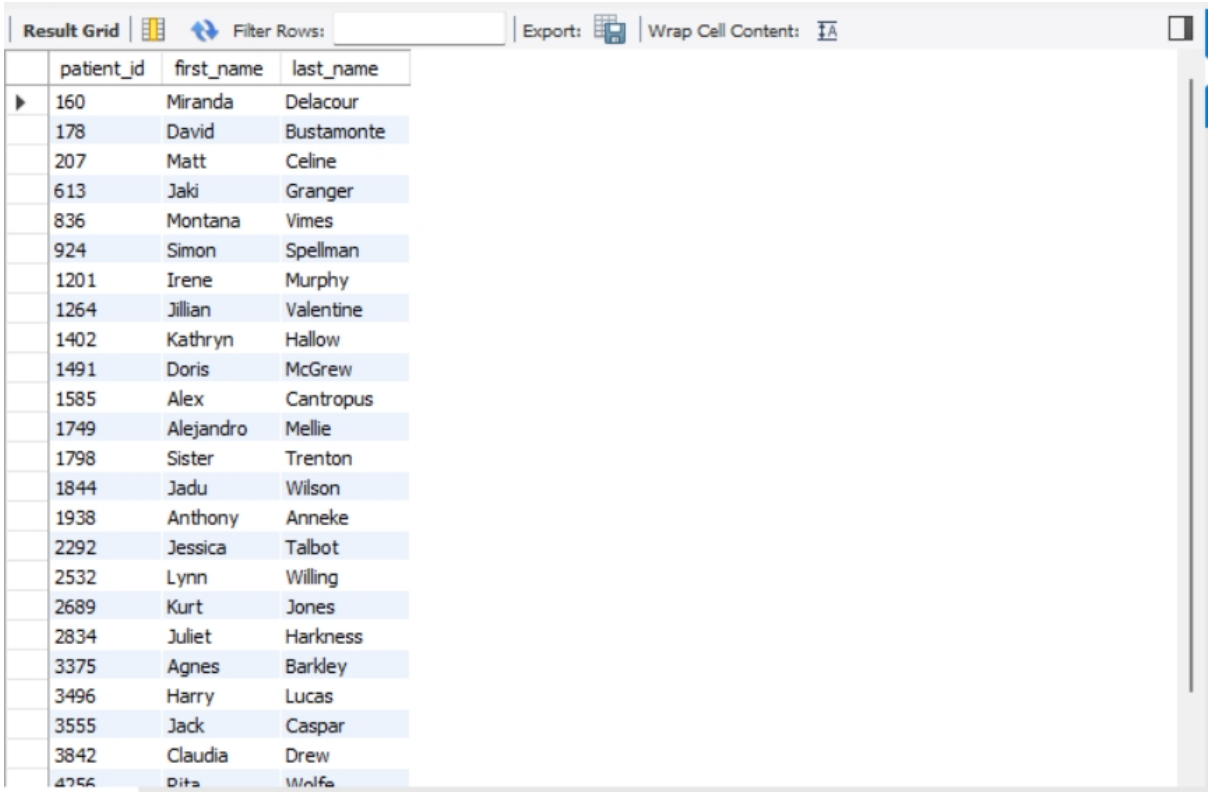
18. Show patient_id and first_name from patients where their first_name start and ends with 's' and is at least 6 characters long.

Q--select patient_id,first_name from patients where first_name like'S%S' and length(first_name)>=6;



19. Show patient_id, first_name, last_name from patients whos diagnosis is 'Dementia'. Primary diagnosis is stored in the admissions table.

Q--select p.patient_id,p.first_name, p.last_name from patients p join admissions a on p.patient_id=a.patient_id where a.diagnosis='Dementia';

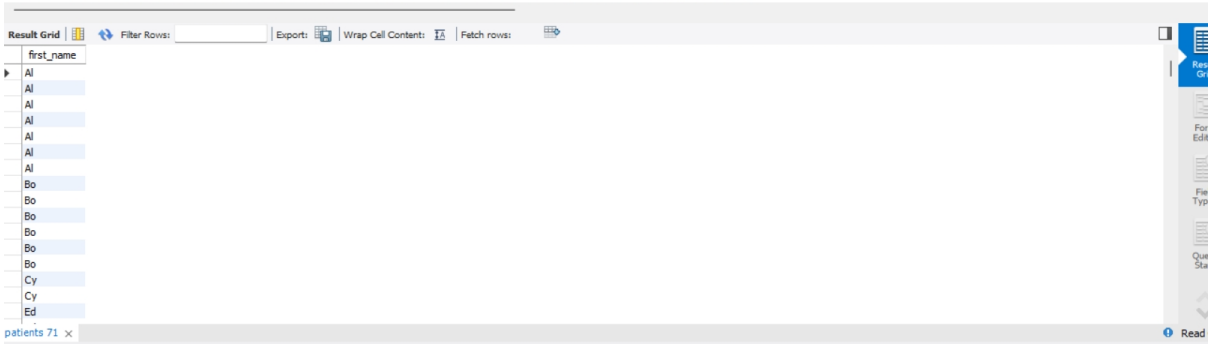


patient_id	first_name	last_name
160	Miranda	Delacour
178	David	Bustamonte
207	Matt	Celine
613	Jaki	Granger
836	Montana	Vimes
924	Simon	Spellman
1201	Irene	Murphy
1264	Jillian	Valentine
1402	Kathryn	Hallow
1491	Doris	McGrew
1585	Alex	Cantropus
1749	Alejandro	Mellie
1798	Sister	Trenton
1844	Jadu	Wilson
1938	Anthony	Anneke
2292	Jessica	Talbot
2532	Lynn	Willing
2689	Kurt	Jones
2834	Juliet	Harkness
3375	Agnes	Barkley
3496	Harry	Lucas
3555	Jack	Caspar
3842	Claudia	Drew

20. Display every patient's first_name. Order the list by the length of each name and then by alphbetically.

Q--select first_name from patients order by length(first_name),first_name;

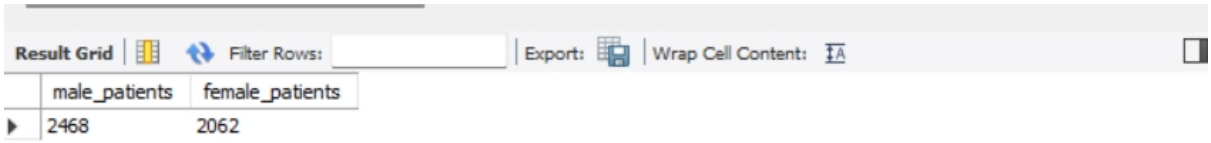
```
39    -- 20 Display every patient's first_name ordered by length then alphabetically
40 •  select first_name from patients order by length(first_name),first_name;
41    -- 21 Show total male and female patients in same row
42 •  select sum(case when gender='M' then 1 else 0 end) as male_patients, sum(case when gender='F' then 1 else 0 end) as female_patients from patients;
43    -- 22 Show total male and female patients in same row
44 •  select sum(case when gender='M' then 1 else 0 end) as male_patients, sum(case when gender='F' then 1 else 0 end) as female_patients from patients;
45    -- 23 Show patient_id, diagnosis for patients admitted multiple times for same diagnosis
46 •  select patient_id,diagnosis from admissions group by patient_id,diagnosis having count(*)>1;
```



first_name
Al
Al
Al
Al
Al
Al
Al
Bo
Bo
Bo
Bo
Bo
Bo
Cy
Cy
Ed

21. Show the total amount of male patients and the total amount of female patients in the patients table. Display the two results in the same row.

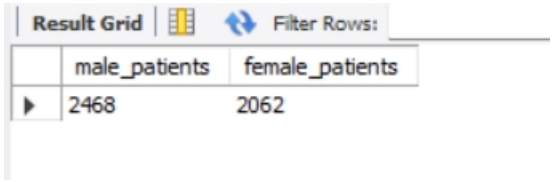
Q--select sum(case when gender='M' then 1 else 0 end) as male_patients, sum(case when gender='F' then 1 else 0 end) as female_patients from patients;



male_patients	female_patients
2468	2062

22. Show the total amount of male patients and the total amount of female patients in the patients table. Display the two results in the same row.

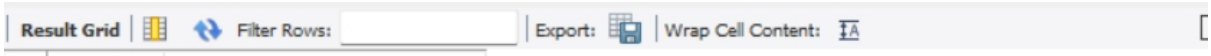
Q2--select sum(case when gender='M' then 1 else 0 end) as male_patients, sum(case when gender='F' then 1 else 0 end) as female_patients from patients;



male_patients	female_patients
2468	2062

23. Show patient_id, diagnosis from admissions. Find patients admitted multiple times for the same diagnosis.

Q3--select patient_id,diagnosis from admissions group by patient_id,diagnosis having count(*)>1;



patient_id	diagnosis
137	Pregnancy
320	Pneumonia
1577	Congestive Heart Failure
2004	Left Shoulder Rotator Cuff Repair
2859	Severed Spine At C3
3012	Appendicitis
3367	Pyelonephritis
3468	Congestive Heart Failure
4083	Congestive Heart Failure
4121	Congestive Heart Failure
4363	Congestive Heart Failure

24. Show the city and the total number of patients in the city. Order from most to least patients and then by city name ascending.

Q--select city,count(*) as total_patients from patients group by city order by total_patients desc,city asc;

```
47    -- 24 Show city and total number of patients ordered by count desc then city asc
48 •  select city,count(*) as total_patients from patients  group by city order by total_patients desc,city asc;
49    -- 25 Show first name, last name, and role for patients and doctors
50 •  select first_name,last_name,'patient' as role  from patients union select first_name,last_name,'doctor' as role from doctors;
51    -- 26 Show all allergies ordered by popularity. Remove NULL values from query.
52 •  select allergies, count(*) as total_diagnosis from patients where allergies is not null group by allergies order by total_diagnosis desc;
53    -- 27 Show all patient's first_name, last_name, and birth_date who were born in the 1970s decade. Sort the list starting from the earliest birth_date.
54 •  select first_name,last_name,birth_date from patients where year(birth_date) between 1970 and 1979 order by birth_date asc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

city	total_patients
Hamilton	1938
Toronto	317
Burlington	276
Brantford	147
Ancaster	117
Stoney Creek	107
Cambridge	79
Dundas	79
Milton	65
Paris	58
Grimsby	55
Timmins	53
Ottawa	51
Halifax	50
Oakville	44
Sarnia	43

Result 72

25. Show first name, last name and role of every person that is either patient or doctor. The roles are either "Patient" or "Doctor"

Q--select first_name,last_name,'patient' as role from patients union select first_name,last_name,'doctor' as role from doctors;

```
49    -- 25 Show first name, last name, and role for patients and doctors
50 •  select first_name,last_name,'patient' as role  from patients union select first_name,last_name,'doctor' as role from doctors;
51    -- 26 Show all allergies ordered by popularity. Remove NULL values from query.
52 •  select allergies, count(*) as total_diagnosis from patients where allergies is not null group by allergies order by total_diagnosis desc;
53    -- 27 Show all patient's first_name, last_name, and birth_date who were born in the 1970s decade. Sort the list starting from the earliest birth_date.
54 •  select first_name,last_name,birth_date from patients where year(birth_date) between 1970 and 1979 order by birth_date asc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

first_name	last_name	role
Donald	Waterfield	patient
Mickey	Baasha	patient
Jiji	Sharma	patient
Blair	Diaz	patient
Charles	Wolfe	patient
Sue	Falcon	patient
Thomas	ONeill	patient
Sonny	Beckett	patient
Sister	Spitzer	patient
Cedric	Coltrane	patient
Hank	Spencer	patient
Sara	di Marco	patient
Daphne	Seabright	patient
Rick	Bennett	patient
Amy	Leela	patient
Woody	Bashir	patient

Result 73

Output

26. Show all allergies ordered by popularity. Remove NULL values from query.

Q--select allergies, count(*) as total_diagnosis from patients where allergies is not null group by allergies order by total_diagnosis desc;

```
51    -- 26 Show all allergies ordered by popularity. Remove NULL values from query.
52 •  select allergies, count(*) as total_diagnosis from patients where allergies is not null group by allergies order by total_diagnosis desc;
53    -- 27 Show all patient's first_name, last_name, and birth_date who were born in the 1970s decade. Sort the list starting from the earliest birth_date.
54 •  select first_name,last_name,birth_date from patients where year(birth_date) between 1970 and 1979 order by birth_date asc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

allergies	total_diagnosis
Penicillin	1082
Codeine	305
Sulfa	157
ASA	99
Sulfa Drugs	71
Peanuts	52
Iodine	48
Tylenol	42
Bee Stings	40
Valporic Acid	38
Tetracycline	34
Wheat	33
Peanut Butter	29
Micropore T...	27
Hay Fever	25
Strawberries	25
Almonds	23
Sulphur	22
Morphine	22
Honey	20

Result 74

27. Show all patient's first_name, last_name, and birth_date who were born in the 1970s decade. Sort the list starting from the earliest birth_date.

Q-select first_name,last_name,birth_date from patients where year(birth_date) between 1970 and 1979 order by birth_date asc;

```
52 •  select allergies, count(*) as total_diagnosis from patients where allergies is not null group by allergies order by total_diagnosis desc;
53    -- 27 Show all patient's first_name, last_name, and birth_date who were born in the 1970s decade. Sort the list starting from the earliest birth_da
54 •  select first_name,last_name,birth_date from patients where year(birth_date) between 1970 and 1979 order by birth_date asc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

first_name	last_name	birth_date
Frances	Kobayakawa	1970-01-02
Sunny	Burrell	1970-01-07
Penelope	Beckett	1970-01-14
Deborah	Stewart	1970-01-14
Augusta	Decker	1970-01-22
Sookie	Brearly	1970-02-01
Temple	Wylie	1970-02-10
Deanna	Spano	1970-03-23
Jadu	Principal	1970-03-28
Betty	Stephens	1970-03-28
Jo	Sahid	1970-03-31
Santino	Hawker	1970-04-08
Andy	Guarnaccia	1970-04-10
Richard	Arroway	1970-04-13
Alan	Davis	1970-04-14
Cameron	Spellman	1970-04-21
Bertha	Smart	1970-05-06
Nicky	Klump	1970-05-19
Rose	Petrie	1970-05-27
Kenny	Skelton	1970-05-29

patients 75

28. We want to display each patient's full name in a single column. Their last_name in all upper letters must appear first, then first_name in all lower case letters. Separate the last_name and first_name with a comma. Order the list by the first_name in decending order EX: SMITH,jane

Q--select concat(upper(last_name),',',lower(first_name)) as full_name from patients order by first_name desc;

```
select first_name,last_name,birth_date from patients where year(birth_date) between 1970 and 1979 order by birth_date asc;
-- 28. We want to display each patient's full name in a single column. Their last_name in all upper letters must appear first, then first_name in all lo
select concat(upper(last_name),',',lower(first_name)) as full_name from patients order by first_name desc;
```

full_name
MILLER,zoe
CORBIE,ziva
KOBAYAKAWA,zenigata
OVERSTREET,zenigata
BENNETT,zen
MEPHESTO,zelda
MORRIS,zelda
THOMAS,zefram
FLUTE,zefram
MARONEY,zatanna
TYRELL,zatanna
CHE,zane
WONG,zane
DREW,zack
BAKSHI,zachary
PRESTON,yusuke
LAZARUS,yuri
PROVENZA,yuri
CHURCHILL,yuri
RANDALL,yuko

29. Show the province_id(s), sum of height; where the total sum of its patient's height is greater than or equal to 7,000.

Q--select province_id,sum(height) as total_height from patients group by province_id having sum(height)>=7000;

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

province_id	total_height
BC	7720
NS	9765
ON	678037

30. Show the difference between the largest weight and smallest weight for patients with the last name 'Maroni'

Q--select max(weight)-min(weight) as weight_diff from patients where last_name='Maroni';

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

weight_diff
71

31. Show all of the days of the month (1-31) and how many admission_dates occurred on that day. Sort by the day with most admissions to least admissions.

Q--select day(admission_date) as day_of_month, count(*) as total_admissions from admissions group by day_of_month order by total_admissions desc;

```
60 •  select max(weight)-min(weight) as weight_diff from patients where last_name='Maroni';
61    -- 31. Show all of the days of the month (1-31) and how many admission_dates occurred on that day. Sort by the day with most admissions to least admissior
62 •  select day(admission_date) as day_of_month, count(*) as total_admissions from admissions group by day_of_month order by total_admissions desc;
63    -- 32. Show all of the patients grouped into weight groups. Show the total amount of patients in each weight group. Order the list by the weight group dec
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

day_of_month	total_admissions
4	184
11	184
9	183
2	180
12	179
6	179
16	177
21	174
13	173
28	173
7	167
8	166
18	166
3	166
20	165
30	165
10	164
15	164
14	164
24	161

Result 78 | Read Only

Output

32. Show all of the patients grouped into weight groups. Show the total amount of patients in each weight group. Order the list by the weight group decending. e.g. if they weight 100 to 109 they are placed in the 100 weight group, 110-119 = 110 weight group, etc.

Q--SELECT FLOOR(weight / 10) * 10 AS weight_group,COUNT(*) AS total_patients FROM patients GROUP BY weight_group ORDER BY weight_group DESC;

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

weight_group	total_patients
140	6
130	59
120	191
110	426
100	507
90	403
80	478
70	633
60	685
50	443
40	202
30	126
20	165
10	114
0	92

33. Show patient_id, weight, height, isObese from the patients table. Display isObese as a boolean 0 or 1. Obese is defined as weight(kg)/(height(m). Weight is in units kg. Height is in units cm.

Q--select patient_id, weight, height, case when weight / power(height / 100.0, 2) >= 30 then 1 else 0 end as isobese from patients;

```
65    -- 33. Show patient_id, weight, height, isObese from the patients table. Display isObese as a boolean 0 or 1. Obese is defined as weight(kg)/(height(m). k
66 •  select patient_id, weight, height, case when weight / power(height / 100.0, 2) >= 30 then 1 else 0 end as isobese from patients;
67    -- 34 Show patient_id, first_name, last_name, and attending doctor's specialty. Show only the patients who has a diagnosis as 'Epilepsy' and the doctor's
68 •  select p.patient_id, p.first_name, p.last_name, d.specialty from patients p join admissions a on p.patient_id = a.patient_id join doctors d on a.attending
69
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

weight_group	total_patients
140	6
130	59
120	191
110	426
100	507
90	403
80	478
70	633
60	685
50	443
40	202
30	126
20	165
10	114
0	92

Result 77 | Read Only

34. Show patient_id, first_name, last_name, and attending doctor's specialty. Show only the patients who has a diagnosis as 'Epilepsy' and the doctor's first name is 'Lisa'. Check patients, admissions, and doctors tables for required information.

Q--SELECT FLOOR(weight / 10) * 10 AS weight_group,COUNT(*) AS total_patients FROM patients GROUP BY weight_group ORDER BY weight_group DESC;

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

patient_id	first_name	last_name	specialty
468	Frank	Anderson	Obstetrician/Gynecologist
701	Precious	Ashton	Obstetrician/Gynecologist